

Robust and Efficient Structure-from-Motion Method for Ambiguous Large-Scale Indoor Scene

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Figure 1: Comparison of reconstruction results of large-scale indoor scene with high ambiguity and repeated structures. (a) Result of COLMAP [Schönberger and Frahm 2016], (b) result of [Su et al. 2020], and (c) result of our method. Both (a) and (b) indicate failure to reconstruct some ambiguous areas and floor collapse. By contrast, our method could successfully construct a complete model, and all parts were connected correctly.

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1 INTRODUCTION

Three-dimensional (3D) reconstruction methods have been used in several fields, such as augmented reality and autonomous navigation. The most famous algorithm is the structure-from-motion (SfM) algorithm. SfM takes two-dimensional images from various viewpoints in the scene as the input, extracts feature points in images, and determines the position of the points and camera poses in 3D space. Although the SfM method has been widely used, it often

fails to reconstruct ambiguous or repeated structures due to false-feature matching. [Cui et al. 2020] argued that the image with the most feature matching is likely the correct image; thus, they used this theory to filter out more outliers, which made their method more robust when used on repeated structures. [Su et al. 2020] first introduced additional inertial measurement unit (IMU) information and successfully reconstructed ambiguous scenes. However, their method cannot be applied in large-scale indoor scenes (Figure 1), which still pose a considerable challenge.

We propose a method for reconstructing a large-scale indoor scene where an accurate viewgraph is constructed with additional IMU information. The results indicate that our method can successfully reconstruct large-scale scenes with highly ambiguous and repeated structures.

2 METHOD

Figure 2 is the overview of our proposed method. Inspired by [Su et al. 2020], we use either additional IMU information or serial numbers to assist the algorithm with distinguishing differences between images. Different from traditional SfM methods, our proposed method requires video sequences rather than individual images to be recorded for ambiguous areas in the scene. Instead of

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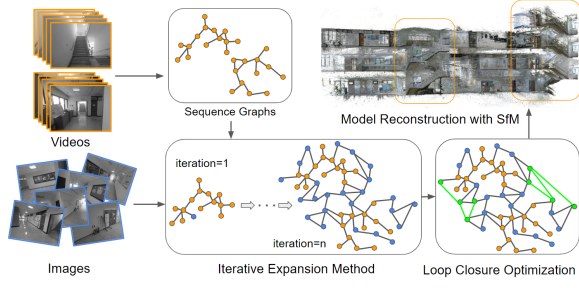


Figure 2: The overview of our proposed method. (a) Build sequence graph for each image sequences, (b) build complete view graph by iteratively adding individual images and connect all sequences graph, (c) reconstruct model with SfM.

using an independent IMU sensor and camera as was done in [Su et al. 2020], we use a smartphone with ARCore to record video and IMU information simultaneously, providing each frame with a relative 3D position. This streamlines the process and rids the camera of calibration or synchronisation problems.

Our view graph reconstruction algorithm includes three steps. First, it will build sub-view graph for each video sequence, we call it sequence graph. Due to the structural ambiguity, we establish some thresholds to filter out outliers. The relative height and drift are used for sequences with IMU information, and the serial number is used as the threshold for sequences without IMU information.

After sequence graphs are reconstructed, we implement an iterative expansion method inspired by [Su et al. 2020] to construct the complete viewgraph G_c . We regard a sequence graph as the root of the complete view graph, and start to expand it iteratively. G_c^j denotes the current G_c at round j . At each iteration, the algorithm collects all nodes that are adjacent to G_c^j as candidate set C . The algorithm always select the target node v_t that has the highest edge weight with a reference node v_r , which already in G_c^j , and add v_t into G_c^j . Instead of calculating a mean score for each candidate as was done in [Su et al. 2020], the algorithm directly chooses the node with the highest edge weight, which is more efficient and yields similarly robust results. To strengthen the relationship between v_t and G_c^j , the algorithm iteratively add edges between v_t and v_r , v_t and v_r 's neighbors, and so on. To avoid adding the incorrect edges between images that should be cut by additional information, v_t inherit additional information from v_r if v_t is an individual image. Thus, the algorithm can reuse the thresholds that were from the previous step to verify each added edge. If v_t belongs to another video sequence, the algorithm add the entire sequence graph to G_c^j . This addresses problems of large-scale scenes with more than one ambiguous structure. Consequently, the method of [Su et al. 2020] was unable to reconstruct large-scale ambiguous scene.

The complete view graph is a tree-like structure due to the expansion strategy. To enhance the robustness of the scene with a loop, we add some edges that are not in trajectories which have sufficient feature-matching and pass geometric verification. Loops in sequences are not considered because they are naturally processed during sequence graph reconstruction. Finally, we use COLMAP [Schönberger and Frahm 2016] to reconstruct the entire scene.



Figure 3: The result of library scene dataset. (a) The result of [Su et al. 2020]'s method, (b) The result of our method.

3 RESULTS AND DISCUSSION

To evaluate our proposed method, we compare the results with COLMAP [Schönberger and Frahm 2016] and [Su et al. 2020] for the *Scene Department_Building* and *Scene Library* dataset. Figure 1 shows the results of *Scene Department_Building*, which is a large-scale indoor scene that includes three aisles on different floors and two sets of stairs, containing 2 video trajectories and about 5k images in total. In Figure 1 (a), COLMAP fails to reconstruct the two stairs due to high ambiguity, and all the aisles are broken. Figure 1 (b) shows that the method of [Su et al. 2020] cannot handle scenes that contain more than one ambiguous structure. Therefore, the method only successfully reconstruct one set of stairs. Our proposed method can successfully reconstruct the entire department building, and all stairs and aisles connect correctly, as shown in Figure 1 (c).

Experiments were also conducted on *Scene Library* to evaluate the generalisability. *Scene Library* is a more general dataset because it includes only one video trajectory and about 3k images. Furthermore, most of the images have wider viewing angles than those in *Scene Department_Building*. Although the dataset is simpler, the method of [Su et al. 2020] cannot reconstruct the scene correctly, as shown in Figure 3 (a). All floors collapse together, and the stairs are severely distorted. By contrast, as shown in Figure 3 (b), our method can successfully reconstruct all three floors and the stairs. Additional experiments can be found in supplementary material.

We proposed a method to reconstruct large-scale ambiguous indoor scene. Different from other previous works, our algorithm can handle scenes that include more than one ambiguous area, and the results indicate that our method is more robust and efficient. Most importantly, none of existing works can successfully construct such ambiguous large-scale indoor scene, to the best of our knowledge.

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